

Maintaining a balance

Vivo's **Keshav Chugh** tells us what it takes to create a great smartphone camera

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To understand what goes around in smartphone camera development, we had a small chat with Vivo's product manager for camera development, Keshav Chugh on the sidelines of Vivo X90 series pre-launch briefing about Vivo's camera development. Having a background in photography, Chugh takes care of Vivo's camera development, which includes coming up with new features, what hardware to use, how to tune the software according to different markets, and much more. Here's how the conversation went:

❏ Please give us an understanding of what goes behind the camera development on Vivo smartphones?

So every time we start with a new device, there is a three-step system that is followed in order to tune the cameras. The first step is coming up with a problem statement. A problem statement is what we want to achieve with a camera, what our focus will be. Then come the optics part, where we select what hardware to use for what device. The optics stage is also where Zeiss steps in. They are the experts in terms of optics and imaging, and they help us tune the hardware according to different markets. The third step is computation, where machine learning and AI play a big part. This is where after having everything in place, we look at the software and how we can use our algorithms to enhance the camera experience.

❏ Can you tell us more about the Vivo V2 chip and what the company's goal with the ISP chip is?



Keshav Chugh
Product Manager, Camera R&D, Vivo

Yes, so apart from handling all the camera processing that goes on in Vivo (and iQOO) flagships, the V2 chip also acts as a Deep Learning Accelerator (DLA) which facilitates and handles all the AI algorithms. The V2 is a scheduling chip that arranges, controls, and optimises workloads in a production process. It basically acts like a manager to help solve problems and implement new camera features.

Ultimately, our goal is not to market V2 or anything that Vivo does, the end goal is that the problem statement should be solved.

❏ Please tell us more about Vivo's partnership with Zeiss?

With Zeiss, we have two types of collaborations - one is co-engineering, one is collaboration. The co-engineering part is where they work with us product-by-product. For example, how they helped us with the camera development in

the Vivo X90 series and other previous X-series flagships. That is product engineering, where they are with us every step of the way in the camera development. Next is collaboration, which is a more long-term thing. Here, we are currently doing R&D on what we'll bring in the future. This involves a lot of experimentation and testing. There is a specific lab at Zeiss in Germany, only for development of Vivo products. So here also, we work on a problem statement basis. For example, how do we bring Zeiss Cine Flare to smartphones is a problem statement.

This is another reason why our partnership with Zeiss has been successful. Because if we were to just slap on filters, a big legacy brand like Zeiss wouldn't want to associate with us. It is because they are happy with the final product, we can confidently use their name.

❏ Who is making the process for the V2? Does Vivo have any plans to license these ISP chips?

So the silicon manufacturing is done by silicon manufacturers, but the design and architecture is proprietary to us. Coming to licensing, there are no plans to license this. That is not the idea behind the Vivo V-series chips. This is something that we have invested a lot of money and resources in, and it is our IP.

❏ What is your favourite camera feature in the Vivo X90 series?

In terms of camera, I'd say I like the portrait modes a lot. Of course, Cine Flare is a new one, but overall all the Zeiss portraits are my favourite. So I was a commercial photographer before this and I love street photography. In street photography, 50mm is the perfect focal length. Pair this with f/1.6 aperture, and it becomes an incredible thing for street photography. This time we have a 30% bigger sensor, so portraits have improved drastically over the previous generation. **[d]**